SSC MTS 2024 question paper with answers

12 November 2024 · Shift 3

77 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 13 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 13 questions

Q.1

The average score of 30 students (boys and girls only) in a class test was 16.2 and that of 18 boys in the same test was 16. The average score of girls in the same test was:

(A) 16.7
(B) 16.5
(C) 16.3
(D) 16.1

Q.2

Mihir buys a toy for ₹70 and sells it for ₹90. What is his gain percentage?

(A) 35.62%
(B) 28.57%
(C) 25.25%
(D) 20%

Q.4

Two trains start at the same time from two stations and proceed towards each other at speeds of 120 km/hr and 80 km/hr, respectively. When they meet, it is found that one train has travelled 60 km more than the other. What is the distance (in km) between the two stations?

(A) 268
(B) 387
(C) 300
(D) 452

Q.5

The average salary of all 50 employees including 5 officers of a company is ₹855. If the average salary of the officers is ₹2,550, find the average salary of the remaining staff of the company. (Rounded to the nearest integer.)

(A) ₹657
(B) ₹677
(C) ₹667
(D) ₹637

Q.6

Using the factor method of HCF, which of the following is the HCF of 160 and 256?

(A) 16
(B) 40
(C) 64
(D) 32

Q.8

The marked price of a mobile phone at a showroom is ₹25,000 and is sold at successive discounts of 20%, 15% and 10%. Find its selling price.

(A) ₹15,000
(B) ₹15,300
(C) ₹14,000
(D) ₹14,960

Q.9

Harsha obtained 72, 63 and 81 marks (out of 100) in Maths, Physics and Chemistry, respectively. What is the average of his marks?

(A) 63
(B) 81
(C) 72
(D) 84

Q.10

A sum of money at simple interest amounts to ₹2,500 in three years and ₹3,000 in five years. What is the rate of interest (correct up to two decimal places)?

(A) 15%
(B) 14.25%
(C) 14.29%
(D) 13.56%

Q.11

In an election there were only two candidates, Geeta and Shyama. Geeta received 15% of the total votes and Shyama received 8700 votes. 60% of the voters did not cast their vote to either of the two candidates. Find the total percentage of votes that Shyama received, assuming that all the votes cast were valid.

(A) 24%
(B) 23%
(C) 25%
(D) 22%

Q.13

In a hospital, 60% of the patients are men, 25% of the patients are women, and 15% of the patients are children. If the average age of the men is 40 years, that of the women is 36 years, and that of the children is 8 years. What is the average age of all the patients?

(A) 34.2 years
(B) 32.4 years
(C) 23.4 years
(D) 24.3 years

Q.14

A policeman travelling at a speed of 120 km/h chases a thief 5000 m away from him, and travelling at a speed of 100 km/h. What is the time taken by the policeman to catch the thief?

(A) 13.5 min
(B) 10 min
(C) 15 min
(D) 20 min

Q.17

Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 6% per year rate of interest for 6 years.

(A) 1300
(B) 1220
(C) 1260
(D) 1200

Q.19

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 3% per year rate of interest for 3 years.

(A) 165
(B) 185
(C) 265
(D) 225

Reasoning Ability and Problem Solving 16 questions

Q.1

TW21 is related to ZC66 in a certain way. In the same way, UX33 is related to AD78. To which of the following is MP45 related, following the same logic?

(A) VS80
(B) VS90
(C) SV80
(D) SV90

Q.2

What should come in place of the question mark (?) in the given series based on the English alphabetical order? TCU, UFP ,VIK, WLF ,?

(A) XOB
(B) XPA
(C) XOC
(D) XOA

Q.4

11 is related to 78 following a certain logic. Following the same logic, 6 is related to 43. To which of the following is 31 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying, etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 159
(B) 249
(C) 145
(D) 218

Q.6

In a code language, 'VGAL' is coded as ‘BEGJ’ and 'XQEB' is coded as ‘DOKZ’. How will 'HTSO' be coded in that language?

(A) MRZL
(B) MSZK
(C) NQYL
(D) NRYM

Q.9

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed) (18, 108, 12) (33, 198, 22)

(A) (40, 252, 28)
(B) (42, 252, 28)
(C) (42, 252, 26)
(D) (42, 254, 28)

Q.10

BZ7 is related to GE15 in a certain way. In the same way, PN13 is related to US21. To which of the following is IG12 related following the same logic?

(A) NL21
(B) PN21
(C) PM20
(D) NL20

Q.11

What should come in place of the question mark (?) in the given series? 6 16 27 39 52 ?

(A) 66
(B) 67
(C) 65
(D) 64

Q.12

In a certain code language, ‘let us go’ is coded as ‘ck ru tb’ and ‘please go away’ is coded as ‘tb mg du’. How is ‘go’ coded in that code language?

(A) mg
(B) tb
(C) ru
(D) du

Q.13

What will come in place of the question mark (?) in the following equation, if '–' and ' + ' are interchanged and '÷' and '×' are interchanged? 15 – 4 + 9 × 3 ÷ 6 = ?

(A) 12
(B) 10
(C) 18
(D) 1

Q.14

Seven executives of a law firm (P, Q, R, S, T, U and V) are sitting at a circular table for a meeting, everyone facing towards the center(but not necessarily in the same order); P is an immediate neighbor of both U and S; Q is sitting to the immediate right of S, and is one of the immediate neighbors of T; V is sitting second to the right of T. Who is sitting second to the left of U?

(A) R
(B) V
(C) Q
(D) T

Q.15

In a certain code language, 'gre pul sep' means 'eat the sofa', 'pul pra sem' means 'clear my sofa' and 'gre boi nad' means 'the great river'. What is the code for 'the' in the same code language?

(A) nad
(B) sep
(C) gre
(D) pul

Q.16

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: No sunflower is a lily. No lily is a rose. Conclusions: (I) No sunflower is a rose. (II) No rose is a lily.

(A) Only conclusion II follows
(B) Neither conclusion I nor II follows
(C) Only conclusion I follows
(D) Both conclusion I and II follow

Q.17

In a code language, 'KITE' is written as 'PRGV' and 'MARBLE' is written as 'NZIYOV'. How will 'DOLPHIN' be written in the same language?

(A) WLOKSRM
(B) XLOKSRM
(C) WMOKSRM
(D) XMOKSRM

Q.18

E, F, G, H, P, S, and T are sitting around a circular table, facing the centre (but not necessarily in the same order). F sits second to the right of T. P sits to the immediate left of S. Only one person sits between P and G when counted from the left of P. H sits third to the right of E.How many people are sitting between H and P when counted from the right of H?

(A) Two
(B) One
(C) Three
(D) None

Q.19

What will come in place of the question mark (?) of the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 7 + 8 – 24 × 6 ÷ 3 = ?

(A) 15
(B) 11
(C) 8
(D) 16

Q.20

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All schools are colleges. No college is a university. Conclusions: (I): No school is a university. (II): All colleges are schools.

(A) Neither conclusion I nor II follows.
(B) Both conclusions I and II follow.
(C) Only conclusion I follows.
(D) Only conclusion II follows.

General Awareness 25 questions

Q.1

The rock-cut caves in Karnataka, mainly at Badami and Aihole, were built under the patronage of the _____.

(A) Cholas
(B) Guptas
(C) Chalukyas
(D) Mauryas

Q.2

Bharatanatyam dancer _____ received the Natya Kala Sarathy award from Sri Parthasarathy Swami Sabha in 2023 at Chennai.

(A) Vidhya Subramanian
(B) Srekala Bharath
(C) Hema Rajagopalan
(D) Sreedevi Unni

Q.3

The Kakrapar Atomic Power Plant is located in which state?

(A) Tamil Nadu
(B) Rajasthan
(C) Maharashtra
(D) Gujarat

Q.4

What is the term 'ace' commonly associated within sports?

(A) A perfect shot in golf
(B) A high jump technique in athletics
(C) A fast serve in tennis
(D) Scoring a goal in soccer

Q.5

A factory set up in my area is causing huge deforestation. Which of the following can we expect to see in the near future? A) Lowering of ground water B) Lower temperatures C) Frequent rain D) Arrival of new animals

(A) A
(B) B
(C) D
(D) C

Q.6

Which of the following statements about the Ozone layer are true? Statement A: It is found in the stratosphere. Statement B: Chlorofluorocarbons (CFCs) are responsible for ozone hole. Statement C: Ozone layer is found around 15–30 km above the earth's surface. Statement D: Ozone layer is expected not to recover anymore now.

(A) Only statements A and D
(B) Only statements A, B and C
(C) Only statements A and B
(D) Only statement B, C and D

Q.7

Which of the following types of volcanic eruptions of Earth's volcanoes are most explosive?

(A) Dykes
(B) Caldera
(C) Flood basalt provinces
(D) Lapolith

Q.8
Which book has been authored by Shovana Narayan and Geetika Kalha?

(A) Indian Mythology through Classical Dance
(B) Cultural Rhythms in Emotions
(C) Dancing in the Family
(D) Dance Gestures: Mirror of Expressions

Q.9
How many Pillar Edicts or Major Pillar Edicts (PEs) of Ashoka are found?

(A) Two
(B) Seven
(C) Fourteen
(D) Nine

Q.10
A famous organisation Self-Employed Women's Association (SEWA) is related with which of the following states?

(A) Gujarat
(B) Haryana
(C) Madhya Pradesh
(D) Uttar Pradesh

Q.11
At the _____, Neeraj Chopra became the first Indian to win a gold medal in javelin throw.

(A) Asian Games, 2022
(B) Asian Games, 2010
(C) Asian Games, 2018
(D) Asian Games, 2014

Q.12
Who among the following is the author of the book, 'A History of British India'?

(A) Bentham
(B) James Mill
(C) Renley
(D) Grenville Austin

Q.13
According to Article 280 of the Indian Constitution, the Finance Commission needs to be appointed for every _____.

(A) one year
(B) three years
(C) five years
(D) two years

Q.14
Lokrang, a five-day festival organised by the Madhya Pradesh Adivasi Lok Kala Academy, begins on _____ every year.

(A) 22nd March
(B) 26th January
(C) 24th July
(D) 26th September

Q.15
A four-carbon chain with a triple bond is called _____.

(A) butyne
(B) butanal
(C) butanol
(D) butene

Q.16
In May 2023, who was sworn in as Central Vigilance Commissioner?

(A) Praveen Kumar Srivastava
(B) Samuel Joseph
(C) Ram Manohar Singh
(D) Dayal Yadav

Q.17
Oak and chestnut trees are found in which type of forest in India?

(A) Mangrove forests
(B) Thorn forests
(C) Montane forests
(D) Tropical deciduous forests

Q.18
In which of the following sessions of Congress did the Party issue a call to celebrate 26th January as 'Purna Swaraj Day'?

(A) Delhi 1923
(B) Karachi 1931
(C) Lahore 1929
(D) Madras 1927

Q.19
Vani Jairam, who won Padma Bhushan 2023, belongs to which state?

(A) Tamil Nadu
(B) Maharashtra
(C) Madhya Pradesh
(D) Gujarat

Q.20
Which Article of the Constitution of India directs the State to provide public assistance to its citizens in case of unemployment, old age, sickness and disablement, within the limits of its economic capacity and development?

(A) Article 32
(B) Article 41
(C) Article 12
(D) Article 20

Q.21
At the time of its adoption, the Constitution contained 395 Articles and _____ Schedules.

(A) 8
(B) 7
(C) 9
(D) 11

Q.22
The last Mughal emperor of India was:

(A) Aurangzeb
(B) Shah Jahan
(C) Bahadur Shah II
(D) Akbar II

Q.23
In which year was the first Socio Economic and Caste Census (SECC) conducted in India?

(A) 1901
(B) 2011
(C) 2001
(D) 1911

Q.24
In 2023, Government of India has announced to set up a/an _____ DigiLocker for use by MSMEs, large businesses and charitable trusts to store and share documents online securely.

(A) Bond
(B) Company
(C) Entity
(D) Macro

Q.25
Padma Subrahmanyam is associated with which dance form?

(A) Bharatanatyam
(B) Manipuri
(C) Kathak
(D) Chhau

English Language and Comprehension 23 questions

Q.1
Choose the correctly spelt word to fill in the blank meaningfully. The citizens of Iraq wanted _____ after years of war.

(A) pease
(B) piece
(C) peace
(D) peice

Q.2

Select the most appropriate meaning of the underlined word. The Prime Minister will “address” the nation tomorrow.

(A) A postal code for sending letters
(B) To deliver a speech
(C) To greet people
(D) A landmark

Q.3

In the following sentence the underlined part contains as error. Alternatives to the underlined part are given as options. Select the correct alternative. Their chance of winning is “slimmest”.

(A) slimly
(B) slimmer
(C) most slim
(D) slim

Q.5

Select the most appropriate option to substitute the underlined segment in the given sentence. Shirin “came off with a brilliant idea” to solve the problem.

(A) came off to a brilliant idea
(B) came up with a brilliant idea
(C) came up in a brilliant idea
(D) came off on a brilliant idea

Q.6

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in the blanks. Martin Luther King Jr. started a crusade for discrimination _____ African Americans. He relied on peaceful demonstrations and demonstrators. His most famous march was a powerful rally of about 250,000 people in Washington in 1963, where he _____ one of the most famous speeches in history.

(A) opposed; dropped
(B) versus; brought
(C) counter; dispatched
(D) against; delivered

Q.7

Select the most appropriate option to fill in the blanks. The teacher asked the students to remain ____, but the noise continued to ____. She finally had to raise her voice to gain control.

(A) silent; ascend
(B) quit; rise
(C) still; elevate
(D) calm; escalate

Q.8

Select the most appropriate meaning of the idiom from the options given below. To accept the gauntlet

(A) To suffer a great insult
(B) To accept failure
(C) To suffer financial loss
(D) To accept a challenge

Q.9

In the following sentence the underlined part contains as error. Alternatives to the underlined part are given as options. Select the correct alternative. Pankaj is a “better” cook.

(A) goodest
(B) best
(C) good
(D) more good

Q.10

Select the correct spelling of the underlined word in the given sentence. Is he ready to risk his “acredation” for the greater good?

(A) accredation
(B) accreditation
(C) acreditation
(D) accraditation

Q.11

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. I still / do not understand / why he did not / reply my email.

(A) I still
(B) reply my email
(C) why he did not
(D) do not understand

Q.12

Select the most appropriate option to fill in the blank. You need to rest the _____ before baking the bread.

(A) paste
(B) dough
(C) lard
(D) doe

Q.13

Select the most appropriate option to fill in the blank. The team captain inquired of his teammates _____ they preferred to choose batting as their initial option.

(A) whether
(B) whither
(C) wither
(D) weather

Q.14

Select the INCORRECTLY spelt word from the given sentence. The teacher's encouragment motivated the students to excel in their studies.

(A) Teacher's
(B) Motivated
(C) Excel
(D) Encouragment

Q.15

Select the most appropriate ANTONYM of the given word. Inferior

(A) Beautiful
(B) Substandard
(C) Superior
(D) Mediocre

Q.16

Read the following sentence and select its future tense form from the options given below: Rachel and Rahul joined forces on a project and successfully completed it within a month.

(A) Rachel and Rahul will join forces on a project and successfully complete it within a month.
(B) Rachel and Rahul have joined forces on a project and successfully completed it within a month.
(C) Rachel and Rahul should join forces on a project and successfully complete it within a month.
(D) Rachel and Rahul are joining forces on a project and successfully completing it within a month.

Q.17

Select the most appropriate synonym of the given word. Hire

(A) Endure
(B) Engross
(C) Enhance
(D) Engage

Q.19

Select the grammatically correct sentence.

(A) A secondary section of the school is located at an end of the street.
(B) The secondary section of a school is located at the end of a street.
(C) The secondary section of a school is located at an end of the street.
(D) The secondary section of the school is located at the end of the street.

Q.20

Select the grammatically correct sentence from the given options.

(A) She was not allowed to enter the hotel as her clothes were dirty.

(B) She was not allowed to enter the hotel as her clothes were dirtier.

(C) She was not allowed to enter the hotel as her clothes was dirtier.

(D) She was not allow to enter the hotel as her clothes were dirty.

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Employers, nowadays, are (1)_____ on team-building as it proves to be useful in being more productive. So, they create occasions for the employees to spend time together. Some employees are sick (2) _____ office festival parties with mandatory dress codes and fun competitions. Some others (3)____ of having to forcibly bond with colleagues during after-work happy hours when they would rather be at home. A French court (4)____ recently that one very much has the right to be boring at the workplace. Seven years after Mr. T was fired from his company for not participating in fun activities, which included going out for drinks after work once a week, the court said that Mr. T was merely exercising his 'freedom of expression'. It added that the company's culture of fun (5)_____ Mr. T 's fundamental right to 'dignity and respect of private life'. However, employers and HR personnel insist that team-bonding initiatives are crucial to the workplace and the employees too feel the same if they want promotions.

Select the most appropriate option to fill in blank number 1.

(A) focuses

(B) focus

(C) focusing

(D) has focused

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Employers, nowadays, are (1)_____ on team-building as it proves to be useful in being more productive. So, they create occasions for the employees to spend time together. Some employees are sick (2) _____ office festival parties with mandatory dress codes and fun competitions. Some others (3)____ of having to forcibly bond with colleagues during after-work happy hours when they would rather be at home. A French court (4)____ recently that one very much has the right to be boring at the workplace. Seven years after Mr. T was fired from his company for not participating in fun activities, which included going out for drinks after work once a week, the court said that Mr. T was merely exercising his 'freedom of expression'. It added that the company's culture of fun (5)_____ Mr. T 's fundamental right to 'dignity and respect of private life'. However, employers and HR personnel insist that team-bonding initiatives are crucial to the workplace and the employees too feel the same if they want promotions.

Select the most appropriate option to fill in blank number 2.

(A) of

(B) with

(C) for

(D) from

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Employers, nowadays, are (1)_____ on team-building as it proves to be useful in being more productive. So, they create occasions for the employees to spend time together. Some employees are sick (2) _____ office festival parties with mandatory dress codes and fun competitions. Some others (3)____ of having to forcibly bond with colleagues during after-work happy hours when they would rather be at home. A French court (4)____ recently that one very much has the right to be boring at the workplace. Seven years after Mr. T was fired from his company for not participating in fun activities, which included going out for drinks after work once a week, the court said that Mr. T was merely exercising his 'freedom of expression'. It added that the company's culture of fun (5)_____ Mr. T 's fundamental right to 'dignity and respect of private life'. However, employers and HR personnel insist that team-bonding initiatives are crucial to the workplace and the employees too feel the same if they want promotions.

Select the most appropriate option to fill in blank number 3.

(A) tired
(B) has tired
(C) have tired
(D) are tired

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Employers, nowadays, are (1)_____ on team-building as it proves to be useful in being more productive. So, they create occasions for the employees to spend time together. Some employees are sick (2) _____ office festival parties with mandatory dress codes and fun competitions. Some others (3)____ of having to forcibly bond with colleagues during after-work happy hours when they would rather be at home. A French court (4)____ recently that one very much has the right to be boring at the workplace. Seven years after Mr. T was fired from his company for not participating in fun activities, which included going out for drinks after work once a week, the court said that Mr. T was merely exercising his 'freedom of expression'. It added that the company's culture of fun (5)_____ Mr. T 's fundamental right to 'dignity and respect of private life'. However, employers and HR personnel insist that team-bonding initiatives are crucial to the workplace and the employees too feel the same if they want promotions.

Select the most appropriate option to fill in blank number 4.

(A) was ruled
(B) have ruled
(C) has ruled
(D) is ruled

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Employers, nowadays, are (1)_____ on team-building as it proves to be useful in being more productive. So, they create occasions for the employees to spend time together. Some employees are sick (2) _____ office festival parties with mandatory dress codes and fun competitions. Some others (3)____ of having to forcibly bond with colleagues during after-work happy hours when they would rather be at home. A French court (4)____ recently that one very much has the right to be boring at the workplace. Seven years after Mr. T was fired from his company for not participating in fun activities, which included going out for drinks after work once a week, the court said that Mr. T was merely exercising his 'freedom of expression'. It added that the company's culture of fun (5)_____ Mr. T 's fundamental right to 'dignity and respect of private life'. However, employers and HR personnel insist that team-bonding initiatives are crucial to the workplace and the employees too feel the same if they want promotions.

Select the most appropriate option to fill in blank number 5.

(A) violate
(B) is violated
(C) will violate
(D) violated

Answer key

Numerical and Mathematical Ability

Q.1 **(B)**	Q.2 **(B)**	Q.4 **(C)**	Q.5 **(C)**	Q.6 **(D)**	Q.8 **(B)**
Q.9 **(C)**	Q.10 **(C)**	Q.11 **(C)**	Q.13 **(A)**	Q.14 **(C)**	Q.17 **(C)**
Q.19 **(D)**					

Reasoning Ability and Problem Solving

Q.1 **(D)**	Q.2 **(D)**	Q.4 **(D)**	Q.6 **(D)**	Q.9 **(B)**	Q.10 **(D)**
Q.11 **(A)**	Q.12 **(B)**	Q.13 **(D)**	Q.14 **(A)**	Q.15 **(C)**	Q.16 **(A)**
Q.17 **(A)**	Q.18 **(D)**	Q.19 **(B)**	Q.20 **(C)**		

General Awareness

Q.1 **(C)**	Q.2 **(B)**	Q.3 **(D)**	Q.4 **(C)**	Q.5 **(A)**	Q.6 **(B)**
Q.7 **(B)**	Q.8 **(A)**	Q.9 **(B)**	Q.10 **(A)**	Q.11 **(C)**	Q.12 **(B)**
Q.13 **(C)**	Q.14 **(B)**	Q.15 **(A)**	Q.16 **(A)**	Q.17 **(C)**	Q.18 **(C)**
Q.19 **(A)**	Q.20 **(B)**	Q.21 **(A)**	Q.22 **(C)**	Q.23 **(B)**	Q.24 **(C)**
Q.25 **(A)**					

English Language and Comprehension

Q.1 **(C)**	Q.2 **(B)**	Q.3 **(D)**	Q.5 **(B)**	Q.6 **(D)**	Q.7 **(B)**
Q.8 **(D)**	Q.9 **(C)**	Q.10 **(B)**	Q.11 **(B)**	Q.12 **(B)**	Q.13 **(A)**
Q.14 **(D)**	Q.15 **(C)**	Q.16 **(A)**	Q.17 **(D)**	Q.19 **(D)**	Q.20 **(A)**
Q.21 **(C)**	Q.22 **(A)**	Q.23 **(D)**	Q.24 **(C)**	Q.25 **(D)**	

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (B) 16.5

All 30 scored 30 × 16.2 = 486 and the 18 boys 18 × 16 = 288. The 12 girls scored 198, an average of 198/12 = 16.5.

Q.2 · Answer (B) 28.57%

Gain = 90 − 70 = ₹20 on a cost of ₹70: 20/70 × 100 = 28.57%.

Q.4 · Answer (C) 300

Each hour the faster train covers 120 − 80 = 40 km more than the other, so they meet after 60/40 = 1.5 hours. Together they cover (120 + 80) × 1.5 = 300 km.

Q.5 · Answer (C) ₹667

All 50 earn 50 × 855 = ₹42,750 and the officers 5 × 2,550 = ₹12,750. The other 45 earn ₹30,000, an average of ₹666.67 ≈ ₹667.

Q.6 · Answer (D) 32

160 = 2^5 × 5 and 256 = 2^8, so the HCF is 2^5 = 32.

Q.8 · Answer (B) ₹15,300

₹25,000 × 0.8 × 0.85 × 0.9 = ₹15,300.

Q.9 · Answer (C) 72

(72 + 63 + 81)/3 = 216/3 = 72.

Q.10 · Answer (C) 14.29%

The two extra years earn 3,000 − 2,500 = ₹500, so ₹250 a year. The principal is 2,500 − 3 × 250 = ₹1,750 and the rate 250/1,750 × 100 = 14.29%.

Q.11 · Answer (C) 25%

Geeta got 15% and 60% voted for neither, so Shyama got 100 − 15 − 60 = 25%.

Q.13 · Answer (A) 34.2 years

Weighted average: 0.6 × 40 + 0.25 × 36 + 0.15 × 8 = 24 + 9 + 1.2 = 34.2 years.

Q.14 · Answer (C) 15 min

The policeman gains 120 − 100 = 20 km/h on the thief, so he closes the 5 km in 5/20 hour = 15 minutes.

Q.17 · Answer (C) 1260

SI = 3500 × 6 × 6/100 = ₹1260.

Q.19 · Answer (D) 225

SI = 2500 × 3 × 3/100 = ₹225.

Reasoning Ability and Problem Solving

Q.1 · Answer (D) SV90

Each letter moves 6 places forward and the number rises by 45: T → Z, W → C, 21 + 45 = 66; U → A, X → D, 33 + 45 = 78. So M → S, P → V, 45 + 45 = 90: SV90.

Q.2 · Answer (D) XOA

First letters move forward 1 (T, U, V, W, so next X); second letters forward 3 (C, F, I, L, so O); third letters back 5 (U, P, K, F, so A): XOA.

Q.4 · Answer (D) 218

The second number is 7 times the first plus 1: 11 × 7 + 1 = 78 and 6 × 7 + 1 = 43. Likewise 31 × 7 + 1 = 218.

Q.6 · Answer (D) NRYM

The letters move +6, −2, +6, −2: VGAL → BEGJ and XQEB → DOKZ. So H → N, T → R, S → Y, O → M: NRYM.

Q.9 · Answer (B) (42, 252, 28)
In each set the first number × 6 and the third × 9 both give the second: 18 × 6 = 108 = 12 × 9 and 33 × 6 = 198 = 22 × 9. Likewise 42 × 6 = 252 = 28 × 9.

Q.10 · Answer (D) NL20
Each letter moves 5 places forward and the number rises by 8: B → G, Z → E, 7 + 8 = 15; P → U, N → S, 13 + 8 = 21. So I → N, G → L, 12 + 8 = 20: NL20.

Q.11 · Answer (A) 66
The differences are 10, 11, 12, 13, so the next is 14: 52 + 14 = 66.

Q.12 · Answer (B) tb
'go' is the only word in both phrases and 'tb' the only code in both, so go = tb.

Q.13 · Answer (D) 1
After interchanging the signs: 15 + 4 – 9 ÷ 3 × 6 = 15 + 4 – 18 = 1.

Q.14 · Answer (A) R
Facing the centre, a person's left is the clockwise side. Clockwise the seats run S, P, U, V, R, T, Q: P between U and S, Q on S's right next to T, and V second to T's right. Second to the left of U is R.

Q.15 · Answer (C) gre
'sofa' is in the first two sentences and 'pul' in both their codes, so sofa = pul. 'the' is in the first and third, whose only shared code is 'gre': the = gre.

Q.16 · Answer (A) Only conclusion II follows
'No lily is a rose' can be reversed: no rose is a lily (II). Sunflowers and roses are each kept apart from lilies, which says nothing about whether any sunflower is a rose, so I does not follow.

Q.17 · Answer (A) WLOKSRM
Each letter is replaced by its opposite in the alphabet (A ↔ Z, B ↔ Y, ...): K → P, I → R, T → G, E → V. So DOLPHIN → WLOKSRM.

Q.18 · Answer (D) None
Facing the centre, a person's left is the clockwise side. Clockwise the seats run S, P, H, G, F, E, T: P on S's left, G second to P's left, F second to T's right and H third to E's right. P sits immediately to H's right, so no one sits between them.

Q.19 · Answer (B) 11
After interchanging the signs: 7 – 8 + 24 ÷ 6 × 3 = 7 – 8 + 12 = 11.

Q.20 · Answer (C) Only conclusion I follows.
Every school is a college and no college is a university, so no school is a university (I). 'All schools are colleges' cannot be reversed into 'all colleges are schools', so II does not follow.

General Awareness

Q.1 · Answer (C) Chalukyas
The cave temples at Badami and Aihole were cut under the Chalukyas of Badami in the sixth and seventh centuries.

Q.2 · Answer (B) Srekala Bharath
Srekala Bharath, a Bharatanatyam dancer and teacher, received the sabha's Natya Kala Sarathy award on 20 March 2023.

Q.3 · Answer (D) Gujarat
The Kakrapar Atomic Power Station stands on the Tapi river in south Gujarat.

Q.4 · Answer (C) A fast serve in tennis
SSC's key gives the tennis serve: an ace is a serve the receiver cannot touch, usually a fast one, and it is the sense the word most commonly has in sport. (Golfers also call a hole in one an ace, which 'a perfect shot in golf' could describe.)

Q.5 · Answer (A) A
Without trees, rainwater runs off instead of soaking in, so ground water is not recharged and its level falls. Deforestation tends to raise local temperatures and reduce rainfall, and it drives animals away.

Q.6 · Answer (B) Only statements A, B and C
The ozone layer lies in the stratosphere, roughly 15–30 km up, and CFCs caused the ozone hole. D is false: since the Montreal Protocol the layer is expected to recover over the coming decades.

Q.7 · Answer (B) Caldera
Calderas are the most explosive of Earth's volcanoes: they erupt so violently that they collapse on themselves instead of building a tall cone. Dykes and lopoliths are intrusive forms, and flood basalts are outpourings of very fluid lava.

Q.8 · Answer (A) Indian Mythology through Classical Dance
Kathak dancer Shovana Narayan and Geetika Kalha wrote 'Indian Mythology through Classical Dance' (2023).

Q.9 · Answer (B) Seven
Ashoka issued seven Major Pillar Edicts; all seven appear together only on the Delhi-Topra pillar.

Q.10 · Answer (A) Gujarat
SEWA, a trade union of self-employed women, was founded by Ela Bhatt in Ahmedabad, Gujarat, in 1972.

Q.11 · Answer (C) Asian Games, 2018
Neeraj Chopra won the javelin gold at the 2018 Asian Games in Jakarta, India's first in the event; he won it again at the 2022 Games, held in 2023.

Q.12 · Answer (B) James Mill
James Mill's 'The History of British India' (1817) divided India's past into Hindu, Muslim and British periods.

Q.13 · Answer (C) five years
Under Article 280 the President constitutes a Finance Commission every fifth year, or earlier if the President considers it necessary.

Q.14 · Answer (B) 26th January
Lokrang, held in Bhopal by the Madhya Pradesh Adivasi Lok Kala Academy, opens on Republic Day, 26 January, every year.

Q.15 · Answer (A) butyne
The ending '-yne' marks a carbon-carbon triple bond, so four carbons give butyne. Butene has a double bond, butanol an -OH group and butanal an aldehyde group.

Q.16 · Answer (A) Praveen Kumar Srivastava
Praveen Kumar Srivastava took the oath as Central Vigilance Commissioner before President Droupadi Murmu in May 2023.

Q.17 · Answer (C) Montane forests
In the montane forests, between about 1,000 and 2,000 m, wet temperate forests of evergreen broad-leaved trees such as oak and chestnut predominate.

Q.18 · Answer (C) Lahore 1929
The Lahore session of December 1929, under Jawaharlal Nehru, made Purna Swaraj Congress's goal and called for 26 January 1930 to be celebrated as Independence Day.

Q.19 · Answer (A) Tamil Nadu
Playback singer Vani Jairam, awarded the Padma Bhushan in 2023, was born in Vellore, Tamil Nadu.

Q.20 · Answer (B) Article 41
Article 41, a Directive Principle, asks the State to secure the right to work, to education and to public assistance in unemployment, old age, sickness and disablement.

Q.21 · Answer (A) 8
As adopted in 1949 the Constitution had 395 Articles in 22 Parts, and 8 Schedules.

Q.22 · Answer (C) Bahadur Shah II
Bahadur Shah II (Bahadur Shah Zafar) was the last Mughal emperor; the British exiled him to Rangoon after the revolt of 1857.

Q.23 · Answer (B) 2011
The first Socio Economic and Caste Census was begun in 2011.

Q.24 · Answer (C) Entity
The Union Budget 2023-24 announced an Entity DigiLocker for MSMEs, large businesses and charitable trusts to store and share documents online securely.

Q.25 · Answer (A) Bharatanatyam
Padma Subrahmanyam is a Bharatanatyam dancer and scholar, who developed the Bharata Nrithyam style from the karanas of the Natya Shastra.

English Language and Comprehension

Q.1 · Answer (C) peace
'Peace', the absence of war, is the word, correctly spelt; 'piece' means a part.

Q.2 · Answer (B) To deliver a speech
As a verb, to 'address' a group is to speak formally to it: the Prime Minister will deliver a speech to the nation.

Q.3 · Answer (D) slim
Nothing is being compared, so the positive degree is needed: 'Their chance of winning is slim.'

Q.5 · Answer (B) came up with a brilliant idea
To 'come up with' an idea is to think of it: 'Shirin came up with a brilliant idea'. To 'come off' is to succeed or to become detached.

Q.6 · Answer (D) against; delivered
Discrimination is 'against' a group, and a speech is 'delivered'.

Q.7 · Answer (B) quit; rise
SSC's key gives 'quit; rise', evidently 'quiet; rise' with 'quiet' misprinted: the teacher asked for quiet, but the noise kept rising. ('Calm; escalate' also reads naturally, though 'calm' speaks of composure rather than of the silence the noise is set against.)

Q.8 · Answer (D) To accept a challenge
To accept (or take up) the gauntlet is to accept a challenge; a knight threw down his gauntlet to issue one.

Q.9 · Answer (C) good
Nothing is being compared, so the positive degree is needed: 'Pankaj is a good cook.'

Q.10 · Answer (B) accreditation
The correct spelling is 'accreditation'.

Q.11 · Answer (B) reply my email
'Reply' takes 'to' before its object: 'why he did not reply to my email'.

Q.12 · Answer (B) dough
Bread is baked from 'dough', which is left to rest before baking; 'doe' is a female deer.

Q.13 · Answer (A) whether
'Whether' introduces an indirect question between alternatives; 'whither' means to where, 'wither' to shrivel, and 'weather' is the state of the atmosphere.

Q.14 · Answer (D) Encouragment
The correct spelling is 'encouragement'.

Q.15 · Answer (C) Superior
'Inferior' means lower in quality or rank; its opposite is 'superior'.

Q.16 · Answer (A) Rachel and Rahul will join forces on a project and successfully complete it within a month.
The simple future uses 'will' with the base form of each verb: 'will join ... and successfully complete'.

Q.17 · Answer (D) Engage
To 'hire' someone is to 'engage' them: to employ them for a job.

Q.19 · Answer (D) The secondary section of the school is located at the end of the street.
The section, the school and the end of the street are all particular ones the reader can identify, so each takes 'the'.

Q.20 · Answer (A) She was not allowed to enter the hotel as her clothes were dirty.
Nothing is compared, so the positive 'dirty' is needed, with the plural 'clothes were' and the passive 'was not allowed'.

Q.21 · Answer (C) focusing
'Employers, nowadays, are focusing on team-building': the present continuous, for what is going on now.

Q.22 · Answer (A) of
One is 'sick of' something one has had too much of.

Q.23 · Answer (D) are tired
The plural 'Some others' takes 'are', and 'tired of' matches 'sick of': 'Some others are tired of having to forcibly bond'.

Q.24 · Answer (C) has ruled
The singular 'A French court' takes 'has': 'has ruled recently that ...', in the active voice.

Q.25 · Answer (D) violated
The court 'added' that the culture of fun 'violated' his right: the reported verb follows the past tense of 'added'.

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